



## Incident handler's journal

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| <b>Date:</b><br>17/09/2026 | <b>Entry: 1</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Description                | I analyzed a ransomware security incident affecting a healthcare company. The incident involved attackers gaining access through a phishing attack and deploying ransomware that encrypted critical files. The ransom demand indicated that the attackers were likely motivated by financial gain.                                                                                                                                                                                                                                                                                 |
| Tool(s) used               | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| The 5 W's                  | <b>Who:</b> An organized group of unethical workers<br><b>What:</b> A ransomware security incident<br><b>When:</b> Tuesday 9:00am<br><b>Where:</b> A healthcare company<br><b>Why:</b> The incident happened because unethical hackers were able to access the company's systems using a phishing attack. After gaining access, the attackers launched their ransomware on the company's systems, encrypting critical files. The attackers' motivation appears to be financial because the ransom note they left demanded a large sum of money in exchange for the decryption key. |
| Additional notes           | 1. How could the health care company prevent an incident like this from occurring again?<br>2. Should the company pay the ransom to retrieve the decryption key?                                                                                                                                                                                                                                                                                                                                                                                                                   |

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| <b>Date:</b><br>18/09/2026 | <b>Entry: 2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Description                | I analyzed a phishing incident in which a malicious email was sent to the HR department of a financial services company. The email contained a password-protected file from a suspicious source, which was downloaded and opened by an employee, triggering a security alert. The incident demonstrated how phishing emails can be used to gain unauthorized access to an organization's systems.                                                                                                                                                                                                                                |
| Tool(s) used               | VirusTotal and Playbook                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| The 5 W's                  | <p><b>Who:</b> A threat actor who goes by the name clyde west in the email</p> <p><b>What:</b> A Phishing Attack</p> <p><b>When:</b> 20/07/2022 09:30:14 AM</p> <p><b>Where:</b> A financial services company</p> <p><b>Why:</b> A malicious email was sent to HR applying for a job position. The email contained a file which was password protected from a very suspicious source ip address. The file was downloaded and opened by an employee which then affected the employee's machine and triggered the alert. The attacker's motivation is to gain access to the organisation's systems through the phishing email.</p> |
| Additional notes           | <ol style="list-style-type: none"> <li>1. Did the organization advertise for a vacant position for an engineer?</li> <li>2. Are employees trained well enough to spot phishing attempts?</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                              |

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| <b>Date:</b><br>19/09/2026 | <b>Entry: 3</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Description                | I used Wireshark to analyze a network packet and examine the communication between an internal host and an external host. I identified the IPv4 addresses, source and destination ports, and timestamp of the packet. The analysis showed communication from port 49653 to port 80, which is commonly associated with HTTP traffic, helping me understand the network communication taking place.                                                                                                                                                              |
| Tool(s) used               | Wireshark                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Activity                   | <p>I used Wireshark to analyze a network packet and examine its communication details. The communication is between internal host 172.21.224.2 and external host 142.250.1.139</p> <p>The packet uses IPv4 and contains a source port of 49653 and a destination port of 80. Port 80 is commonly associated with HTTP traffic. I examined the packet details to understand how the source and destination hosts were communicating and to identify the protocols and network information involved. The timestamp for the packet was 2022/11/23 at 12:38:34</p> |
| Additional notes           | The packet uses ipv4 for network-layer communication                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

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| <b>Date:</b><br>19/09/2026 | <b>Entry: 4</b><br>Record the journal entry number.                            |
| Description                | I used the command-line tool tcpdump to capture and analyze network traffic. I |

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|                  | <p>identified the available network interface, captured HTTP traffic on port 80, saved the packets to a PCAP file, and reviewed the captured data. This activity helped me understand how tcpdump can be used to capture and inspect network packets from the command line.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Tool(s) used     | Tcpdump tool                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Activity         | <p>I used the command-line interface and the tcpdump tool to capture and analyze network packets. I first used <code>sudo ifconfig</code> to view the available network interfaces. I then used <code>sudo tcpdump -D</code> to display the interfaces available for packet capture.</p> <p>Next, I used <code>sudo tcpdump -i eth0 -v -c 5</code> to capture five packets from the eth0 interface with verbose output. I then created a more specific packet capture using <code>sudo tcpdump -i eth0 -nn -c 9 port 80 -w capture.pcap &amp;</code>. This command captured nine packets associated with port 80 and saved the captured traffic to a file named capture.pcap.</p> <p>To generate network traffic for the capture, I used <code>curl</code> <a href="https://opensource.google.com">opensource.google.com</a>. I then used <code>ls -l capture.pcap</code> to verify that the capture file was created.</p> <p>Finally, I analyzed the captured packets using <code>sudo tcpdump -nn -r capture.pcap -v</code> to read the packet capture with verbose information. I also used <code>sudo tcpdump -nn -r capture.pcap -X</code> to display the packet contents in hexadecimal and ASCII formats.</p> |
| Additional notes | <p>This activity demonstrated how tcpdump can be used from the command line to identify network interfaces, capture specific network traffic, save packets to a .pcap file, and analyze the captured data. I also learned how filtering traffic by a port, such as port 80, can help focus packet analysis on specific types of network communication.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

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| <b>Date:</b><br>21/09/2026 | <b>Entry: 5</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Description                | I used a phishing incident response playbook to investigate a suspicious email and identify indicators of compromise. I analyzed the sender information, email content, attachment, and file hash, which confirmed the file as malicious. Based on my findings, I escalated the incident to a Level 2 SOC analyst for further investigation and response.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Tool(s) used               | Phishing Incident Response Playbook                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Activity                   | <p>I used a phishing incident response playbook to investigate a phishing alert and determine the appropriate response. I reviewed the alert details, email information, sender information, subject line, message content, and attachment associated with the incident.</p> <p>During the investigation, I identified several indicators of a potential phishing attempt. There was an inconsistency between the sender's email address, the name used in the body of the email, and the sender's displayed name. The email also contained grammatical errors and included a password-protected executable file named bfsvc.exe. The file had been downloaded and opened on the affected machine, and its SHA-256 hash was identified as a known malicious file.</p> <p>Based on the evidence and the medium severity of the alert, I determined that the incident required further investigation. I therefore escalated ticket A-2703 to a Level 2 SOC analyst for additional analysis and response.</p> |
| Additional notes           | The playbook helped me systematically examine the phishing email and identify indicators of compromise, including the suspicious sender information, social-engineering characteristics, malicious attachment, and known malicious                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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|  | file hash. The activity also demonstrated the importance of escalating an incident when the available evidence indicates that further investigation and containment may be required. |
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| <b>Date:</b><br>22/09/2026 | <b>Entry: 6</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Description                | I used Suricata to analyze a packet capture using custom detection rules and identify potentially suspicious network activity. I reviewed the alerts generated in the Suricata logs and used <code>jq</code> to extract important information from the <code>eve.json</code> file, including alert signatures, protocols, and destination IP addresses. This activity helped me understand how signature-based intrusion detection can be used to identify and investigate network threats.                                                                                                                                                                                                                                                                                                                                        |
| Tool(s) used               | Suricata                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Activity                   | <p>I used Suricata to explore how signature-based network intrusion detection can be used to identify potentially suspicious activity. I first reviewed the contents of the custom detection rules and examined the Suricata log directory to understand the available output files.</p> <p>I then used Suricata to analyze a packet capture file (<code>sample.pcap</code>) using the custom rules. I disabled checksum validation for the analysis and examined the resulting log files to determine whether any of the rules generated alerts.</p> <p>I reviewed the <code>fast.log</code> file to examine the alerts generated by Suricata. I also analyzed the <code>eve.json</code> file, which contains structured event information generated by Suricata. Using <code>jq</code>, I filtered the JSON data to make the</p> |

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|                  | information easier to analyze and extracted fields such as the timestamp, flow ID, alert signature, protocol, and destination IP address.                                                                                                                                            |
| Additional notes | This activity demonstrated how Suricata can use signatures and custom rules to detect specific patterns in network traffic. I also learned how Suricata records detection results in log files and how jq can be used to extract relevant fields from JSON logs for easier analysis. |

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| <b>Date:</b> 21/09/2026 | <b>Entry:</b> 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Description             | I reviewed an incident final report describing a data breach that exposed approximately 50,000 customer records. The investigation identified a vulnerability in an e-commerce application that allowed unauthorized access to customer information through a forced browsing attack. Reviewing the report helped me understand how security teams identify root causes, analyze logs, respond to incidents, and recommend measures to prevent future attacks.                                                                                                                                                                                                                                                                                     |
| Tool(s) used            | Incident final report                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Activity                | <p>I reviewed an incident final report to understand how a security incident was investigated, documented, and resolved. The report described a data breach in which an attacker gained unauthorized access to customer personally identifiable information (PII) and financial information, affecting approximately 50,000 customer records.</p> <p>The investigation identified a vulnerability in the organization's e-commerce web application that allowed the attacker to perform a forced browsing attack. By modifying the order number in the URL of a purchase confirmation page, the attacker was able to access customer transaction information. Web server access logs were then analyzed and showed an unusually high volume of</p> |

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|                  | <p>sequential customer order requests, which helped confirm how the data had been accessed.</p> <p>I also reviewed the response and remediation actions taken by the organization. The organization notified affected customers, provided free identity protection services, and recommended additional security measures, including routine vulnerability scanning, penetration testing, and stronger access controls.</p> |
| Additional notes | <p>Reviewing the final report helped me understand how security teams document the complete lifecycle of an incident, from the initial alert and investigation through identification of the root cause, response, remediation, and recommendations for preventing similar incidents in the future.</p>                                                                                                                     |

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| <b>Date:</b><br>24/09/2026 | <b>Entry: 8</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Description                | <p>I analyzed a phishing-related security alert involving a malicious file attachment sent to an employee at a financial services company. The file was detected by the organization's intrusion detection system after the employee downloaded and executed it. The investigation included reviewing the file's SHA-256 hash to identify the malicious attachment and understand the cause of the alert.</p>                                                                                                                                          |
| Tool(s) used               | <p>For this activity, I used VirusTotal, which is an investigative tool that analyzes files and URLs for malicious content such as viruses, worms, trojans, and more. It's a very helpful tool to use if you want to quickly check if an indicator of compromise like a website or file has been reported as malicious by others in the cybersecurity community. For this activity, I used VirusTotal to analyze a file hash, which was reported as malicious.</p> <p>This incident occurred in the Detection and Analysis phase. The scenario put</p> |

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|                  | <p>me in the place of a security analyst at a SOC investigating a suspicious file hash. After the suspicious file was detected by the security systems in place, I had to perform deeper analysis and investigation to determine if the alert signified a real threat.</p>                                                                                                                                                                                                                                                                                                                                                    |
| The 5 W's        | <ul style="list-style-type: none"> <li>• <b>Who:</b> An unknown malicious actor</li> <li>• <b>What:</b> An email sent to an employee contained a malicious file attachment with the SHA-256 file hash of<br/>54e6ea47eb04634d3e87fd7787e2136ccfbcc80ade34f246a12cf93bab527f6b</li> <li>• <b>When:</b> At 1:20 p.m., an alert was sent to the organization's SOC after the intrusion detection system detected the file</li> <li>• <b>Where:</b> An employee's computer at a financial services company</li> <li>• <b>Why:</b> An employee was able to download and execute a malicious file attachment via e-mail.</li> </ul> |
| Additional notes | <p>How can this incident be prevented in the future? Should we consider improving security awareness training so that employees are careful with what they click on?</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

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Reflections/Notes:

**1. Were there any specific activities that were challenging for you? Why or why not?**

I really found the activity using tcpdump challenging. I am not new to using the command line, but learning the syntax for a tool like tcpdump was a big learning curve. At first, I felt very frustrated because I wasn't getting the right output. I redid the activity and figured out where I went wrong. What I learned from this was to carefully read the instructions and work through the process slowly.

**2. Has your understanding of incident detection and response changed after taking this course?**

After taking this course, my understanding of incident detection and response has definitely evolved. At the beginning of the course, I had some basic understanding of what detection and response entailed, but I didn't fully understand the complexity involved. As I progressed through the course, I learned about the lifecycle of an incident; the importance of plans, processes, and people; and tools used. Overall, I feel that my understanding has changed, and I am equipped with more knowledge and understanding about incident detection and response.

**3. Was there a specific tool or concept that you enjoyed the most? Why?**

I really enjoyed learning about network traffic analysis and applying what I learned through network protocol analyzer tools. It was my first time learning about network traffic analysis, so it was both challenging and exciting. I found it really fascinating to be able to use tools to capture network traffic and analyze it in real time. I am definitely more interested in learning more about this topic, and I hope to one day become more proficient in using network protocol analyzer tools.